Hypothesis Testing

Lecture 12

Topics in this lecture

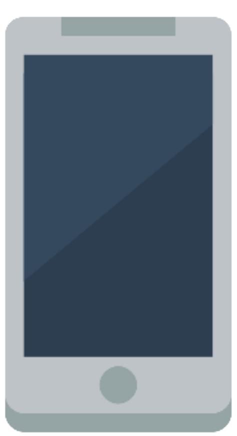
- Large-sample test for a population mean
- Small-sample test for a population mean

Case Study

- A certain type of mobile phone consumes a mean power of 100 mW. A modification to the design is proposed that may reduce power consumption. The new design will be put into production if its mean consumption rate is less than 100 mW.

- A sample of 50 modified phone is built and tested. The sample mean of power consumption rate is 93 mW, and the sample standard deviation is 20 mW.

- Should we put this new design into a production?



x 50

$$\overline{X} = 93$$

$$s = 20$$

Production?

Question to be answered

- Is it plausible that this sample with its mean of 93 mW, could have come from a population whose mean is 100 mW or more?

- Two possible interpretations of observation:

    - The population mean is actually greater than or equal to 100, and the sample mean is lower than this only because of random variation from the population mean. Thus, power consumption will not go down if the new design is used.

    - The population mean is actually less than 100, and the sample mean reflects this fact. Thus, the sample represents a real difference that can be expected if the new design is put into production

Formal Formulation

- Null hypothesis

$$H_0 : \mu \geq 100$$

- Alternate hypothesis

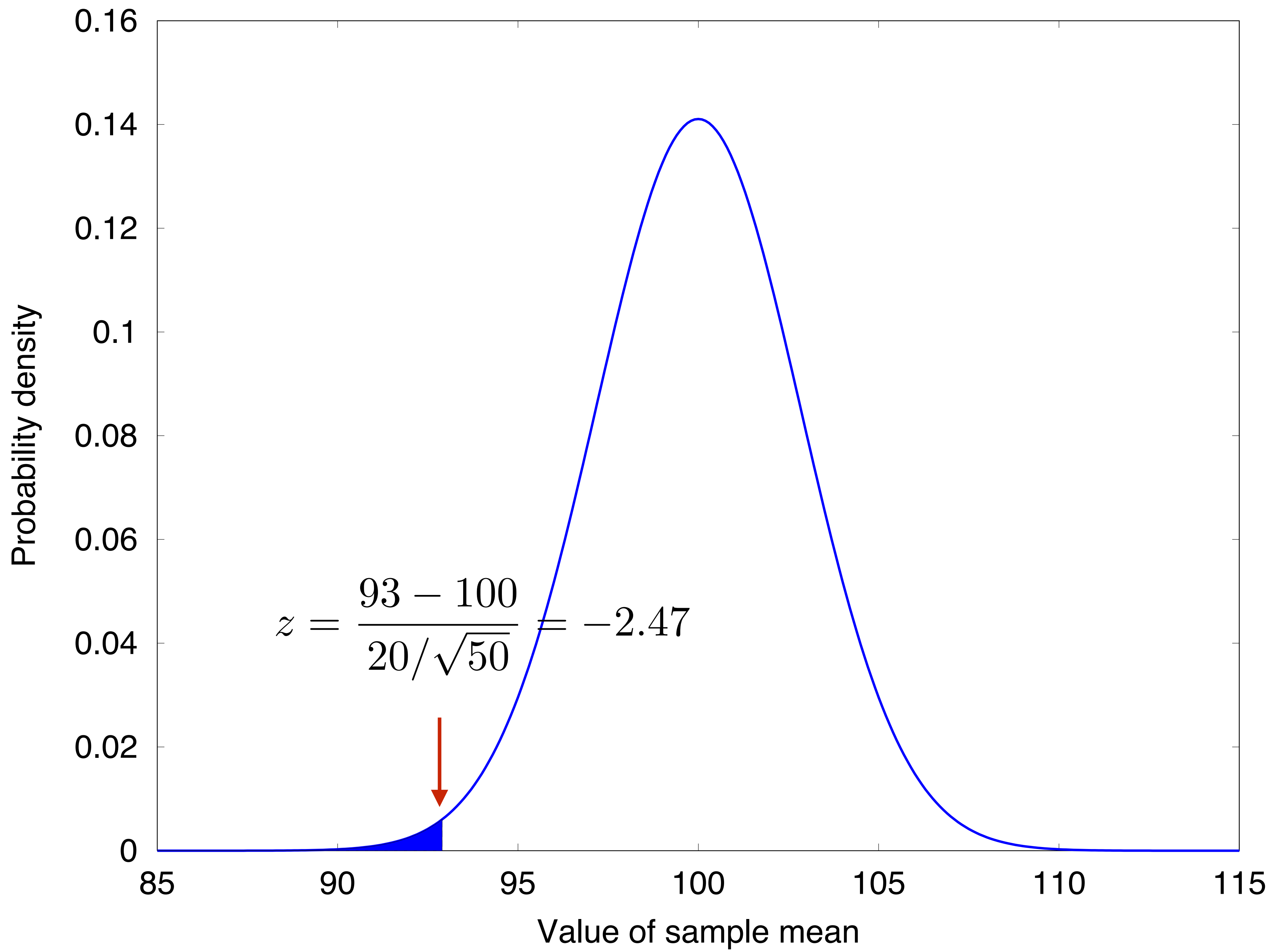
$$H_1 : \mu < 100$$

- Assume that H_0 is true and test if we can reject this hypothesis

If H_0 is true

$$H_0 : \mu \geq 100$$

$$H_1 : \mu < 100$$



P-value

- The hypothesis test involves measuring the strength of the disagreement between the sample and H_0 to produce a number between 0 and 1, called a *P-value*.
- The P-value measures the plausibility of H_0. The smaller the P-value, the stronger the evidence against H_0.
- If P-value is sufficiently small, we maybe willing to reject H_0 and believe H_1 instead.
- Typical rule of thumb: reject H_0 when $P \leq 0.05$

Steps in Hypothesis Test

1. Define H_0 and H_1

2. Assume H_0 to be true

3. Compute a test statistic. A test statistic is a statistic that is used to asses the strength of the evidence against H_0

4. Compute the *P-value* of the test statistic. The P-value is the probability assuming H_0 to be true, that the test statistic would have a value whose disagreement with H_0 is as great as or greater than that actually observed. The P-value is also called the observed *significant level*.

Drawing Conclusion from Hypothesis Test

- The smaller the P-value, the more certain we can be that H_0 is false

- The larger the P-value, the more plausible H_0 becomes, but we can never be certain that H_0 is true.

- A rule of thumb suggests to reject H_0 whenever P-value ≤ 0.05. While this is a rule of convenience, it has no scientific basis

- *Statistical significance*: Whenever the P-value is less than a particular threshold, the result is said to be "statistically significant" at that level. For example, if $P \leq 0.05$, the result is statistically significant at 5% level; if $P \leq 0.01$, the result is statistically significant at 1% level.

Testing Hypothesis

- Assume that μ = 100, and sample standard deviation is s = 20. Thus, $\overline{X}$ has a normal distribution with mean 100 and standard deviation 20/√50 = 2.83.
- The null distribution is $\overline{X} \sim \mathcal{N}\left(100, 2.83^2\right)$
- Compute the P-value for the observed value of $\overline{X} = 93$
  - The corresponding z-score is $z = \dfrac{93 - 100}{20/\sqrt{50}} = -2.47$
  - From the Z-table, we see that the P-value at z = -2.47 is 0.0068

Drawing Conclusion

- The P-value tells us that if H_0 were true, the probability of drawing a sample whose mean was as far from H_0 as the observed value of 93 is only 0.0068. Therefore, one of the following two conclusions is possible

  - H_0 is false

  - H_0 is true, which implies that of all the samples that might have been drawn, only 0.68% of them have a mean as small as or smaller than that of the sample actually drawn. In other words, our sample mean lies in the most extreme 0.68% of its distribution.

Statistical Significance

- Let α be any value between 0 and 1. Then if $P \leq \alpha$
  - The result of the test is said to be statistically significant at the $100\alpha\%$ level
  - The null hypothesis is rejected at the $100\alpha\%$ level.
  - When reporting the result of a hypothesis test, report the P-value, rather than just comparing it to 5% or 1%.

Example 1

- A specification of a safety cable requires that the mean sustainable load should be greater than 120 kg. In an experiment, 60 cables are selected for testing. The sample mean of sustainable load is 123.2 kg, and the sample standard deviation is 8 kg. Should the production of the cables pass the requirement?

Example 2

- A scale is to be calibrated by weighing a 100 g test weight 75 times. The 75 scale readings have mean 100.25 g and standard deviation 1 g. Find the P-value for testing H_0: $\mu = 100$ versus H_1: $\mu \neq 100$

Summary

- Let $X_1, \ldots, X_n$ be a large (e.g., $n > 30$) sample from a population with mean μ and standard deviation σ. To test the hypothesis of the form $H_0: \mu \leq \mu_0$, $H_0: \mu \geq \mu_0$ or $H_0 : \mu = \mu_0$

  - Compute the Z-score: $$z = \frac{\overline{X} - \mu_0}{\sigma / \sqrt{n}}$$

    If σ is unknown it may be approximated with s.

  - Compute the P-value. The P-value is an area under the normal curve, which depends on the alternate hypothesis as follows

Alternate Hypothesis	P-Value
• $H_1: \mu > \mu_0$	Area to the right of z
• $H_1: \mu < \mu_0$	Area to the left of z
• $H_1: \mu \neq \mu_0$	Sum of the areas in the tails cut off by z, -z

Small-sample Test for a Population Mean

- Let $X_1, \ldots, X_n$ be a small (e.g., $n < 30$) sample from a *normal* population with mean μ and standard deviation σ, where σ is unknown. To test the hypothesis of the form H_0: $\mu \leq \mu_0$, H_0: $\mu \geq \mu_0$ or H_0: $\mu = \mu_0$
  - Compute the test statistic: $$t = \frac{\overline{X} - \mu_0}{s/\sqrt{n}}$$
  - Compute the P-value. The P-value is an area under the Student's t curve with n-1 degrees of freedom, which depends on the alternate hypothesis as follows

Alternate Hypothesis	P-Value
• H_1: $\mu > \mu_0$	Area to the right of t
• H_1: $\mu < \mu_0$	Area to the left of t
• H_1: $\mu \neq \mu_0$	Sum of the areas in the tails cut off by t, -t

  - If σ is known, use Z-score as a test statistic, and Z-test should be performed

Example 3

- A battery manufacturer wants to make sure that the mean battery life is greater than 8 hours per full-charge. A sample of 10 batteries are drawn from a population (assuming a normal population). The sample mean is 9 hours and the sample standard deviation is 1 hours. Should the manufacturer claim that the mean battery life is greater than 8 hours per full-charge?

Example 4

- A filling machine is supposed to fill a bottle such that the mean volume is 100 ml. A sample of 5 bottles is drawn and measured. The 5 volumes are 100.03, 100.01, 99.98, 100.2, and 99.95. Assuming that the sample comes from a normal population. Should the machine need calibration?